AI-Powered Smart Agriculture Platform (AgriSmart AI)

Designed for Indian Farmers | Aligned with ICAR & CIBRC Standards

Overview: AgriSmart AI delivers real-time computer vision crop identification, plant leaf disease detection with quantitative chlorophyll/lesion metrics, laboratory soil health card calculations (ICAR NPK dosages), live agro-weather spray windows via Open-Meteo, CIB&RC-registered pesticide recommendations, and a bilingual (English & Telugu) AI assistant.

1. Technology Stack & Multi-Channel Deployment

Layer	Technology Used	Role in Platform
Frontend (Web)	React 18, Vite, Tailwind CSS, Recharts	Responsive cards, charts, Leaflet farm mapping
Backend (API)	Python FastAPI, Uvicorn, SQLite3	REST endpoints, SQLite persistence, schema validation
AI / Computer Vision	PyTorch 2.8, Torchvision, Pillow, NumPy	MobileNetV2 crop classifier & PlantVillage leaf CNN
Cloud App (1-Click)	Streamlit 1.50, Pandas, Altair	Streamlit Cloud public app for mobile & web
Meteorology API	Open-Meteo Live API (Free & Keyless)	Real-time weather, humidity, and safe spray alerts

2. Core Agricultural Capabilities

* **AI Crop Identification:** Classifies 10 major Indian crops (Rice, Wheat, Cotton, Maize, Tomato, Potato, Chilli, Groundnut, Sugarcane, Soybean) with season, duration, temperature, water requirements, and ICAR cultivation best practices.

* **Plant Disease Detection:** Detects Early/Late Blights, Rust, Rice Blast, and Leaf Curl Virus. Accurately flags healthy vs infected tissue with quantitative visual metrics (% chlorophyll, % lesions, severity). Warns if confidence < 70%.

* **Soil Analysis & Nutrients:** Separates visual image observations from laboratory soil test card numbers. Calculates Soil Health Score (0-100) and exact ICAR fertilizer prescriptions: Urea in 3 split doses, DAP/SSP, MOP, and Gypsum/Lime.

* **Location-Based Agro-Weather:** Interactive Leaflet OpenStreetMap with GPS pin & Indian village geocoding. Real-time Open-Meteo weather with actionable Spray Window Alerts (warns against spraying before rain or high wind).

* **CIB&RC Pesticide Advisory:** Official Indian Central Insecticides Board approved active ingredients (Mancozeb 75 WP, Tricyclazole, Imidacloprid, Neem Oil, Trichoderma) with exact 15L knapsack dilution and Pre-Harvest Intervals (PHI).

* **Bilingual AI Assistant:** Chatbot in simple English and Telugu answering common questions on leaf yellowing (chlorosis vs nitrogen deficiency), crop suitability, fertilizer scheduling, and irrigation with Web Speech TTS audio.

* **Farmer Dashboard & My Farm:** KPI cards, Recharts NPK balance chart, farmstead land acreage editor, and SQLite persistent historical diagnostic logs.

3. Deployment Links & Access Commands

Environment / Target	URL / Repository Location	Access Command
GitHub Repository	https://github.com/shaikmohammedshoaib666/AI	git push origin main
Streamlit Cloud App	share.streamlit.io (Auto-detects streamlit_app.py)	1-Click deploy from GitHub
FastAPI Backend (Local)	http://localhost:8000 (Swagger: /docs)	uvicorn main:app --port 8000
React Frontend (Local)	http://localhost:5173	npm run dev
Streamlit App (Local)	http://localhost:8501	streamlit run streamlit_app.py

Statutory Notice: All chemical formulations comply with CIB&RC registers. Always read product container labels before application. Kisan Call Centre Toll-Free: 1800-180-1551 (Available in 22 Indian languages).